

next-door neighbors raised the other twin. Yet, it is difficult to escape the suggestion of at least a small genetic contribution, because it is unlikely that monozygotic twins reared apart would have *more* similar environments than dizygotic twins who grow up in the same household.

Twins and Single Births

One final point about Figure 9.13: Dizygotic twins resemble each other more closely than single-birth siblings do. That finding, consistent across several studies, indicates an influence from being born at the same time, since the genetic similarity between dizygotic twins is the same as that between siblings. These results may reflect the importance of prenatal and early postnatal environment. Twins share a prenatal environment, and therefore get the same nutrition, body temperature, alcohol or tobacco, and all other influences, while the brain is going through some critical stages of development (Devlin, Daniels, & Roeder, 1997). In fact, it is often difficult to distinguish between the role of genetics and that of prenatal environment.

Adopted Children

Another line of evidence comes from studies of adopted children. Children who are reared by their biological parents generally have IQ scores similar to those of their parents (Figure 9.14). For adopted children, IQ scores in early childhood are generally intermediate between those of their biological parents and those of their adoptive parents; as they become older children and then adolescents, their IQ scores gradually become closer to those of their biological parents, and it thus becomes harder to demonstrate the influence of their adoptive parents (Loehlin, Horn, & Willerman, 1989; Plomin, Fulker, Corley, & DeFries,

1997). Furthermore, the IQ scores of unrelated children adopted by the same family resemble each other less than do the IQs of related children who are adopted by separate families (Teasdale & Owen, 1984).

The interpretation of these results is confounded to some extent by the policies of adoption agencies. Many adoption agencies place children of parents with high IQs with the brightest available adoptive parents, thereby providing an environment that should help the children to resemble their biological parents. Still, because adopted children resemble their biological parents *more* than they resemble their adoptive parents, heredity seems to play a significant role in IQ scores.

Gene Identification

The most direct kind of evidence is the identification of specific genes linked to intelligence. One set of investigators compared the chromosomes of children with very high IQ scores (mean 136) and those with average scores (mean 103). They found one gene, of the 37 they examined, that was significantly more common in the high-IQ group (Chorney et al., 1998). Because they had examined so many genes, they knew that a discrepancy in one gene could arise by accident, so they repeated their study with other samples of people with high and average IQs, and again they found that the gene in question was more common in the high-IQ groups. Overall, 46% of the high-IQ people and 23% of the low-IQ people had this gene. Obviously, this gene is neither necessary nor sufficient for high intelligence; it can be only one contributor among dozens of other genes and a great many environmental influences.

Most researchers agree that hereditary and environmental factors both contribute to the observed variations in IQ scores, at least within the European-American population, the subjects of most studies (Thompson, Detterman, & Plomin, 1991; Turkheimer, 1991). The important question is no longer *do* heredity and environment contribute to IQ, but *how* do they contribute? (The one identified gene that appears to be linked to IQ controls a receptor whose functions are very poorly understood.)

Heredity does contribute to variations in IQ scores, but this does not mean that people are somehow limited by the abilities they had at birth. Obviously, if we gave every child either an extremely good or extremely bad environment, we could raise or lower all their IQ scores. Positive heritability of IQ scores merely means that, if children grow up in the same environment, some do better than others, and part of that difference relates to genetics.

A variety of programs have attempted to take children from extremely deprived homes and give them special intervention to improve their intellectual development. People who hoped that a brief intervention might lead to huge, long-term gains have been disappointed. However, intensive programs occupying many hours per week for several years can produce significant, lasting benefits (Ramey & Ramey,

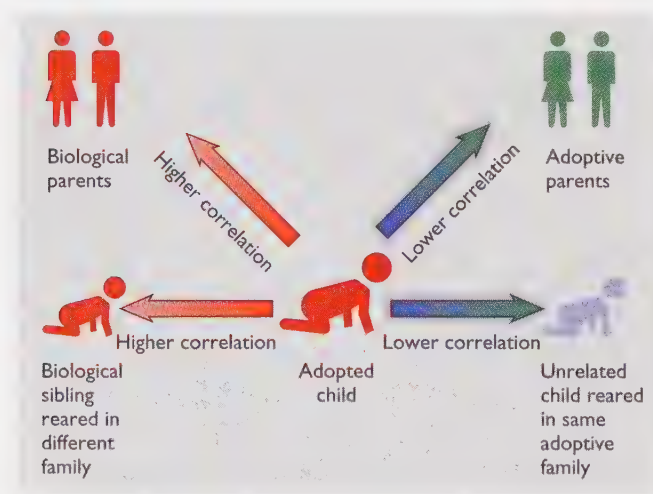


FIGURE 9.14 The IQ scores of adopted children correlate more highly with those of their biological relatives than with those of their adoptive relatives. Such data point to a probable role of heredity.

1998). Again, the point is that conceding that heredity plays a role does not minimize the importance of the environment.

Heredity, Environment, and Ethnic Differences

Is it likely that hereditary differences contribute to part of the observed differences in IQ scores among ethnic groups? Some psychologists believe so but others do not. Many, on each side of this argument, state their case with more confidence than their data would seem to support.

Those who believe that heredity is a likely contributor to ethnic differences offer the following simple proposition: Hereditary differences do appear to contribute to IQ differences within at least one ethnic group (Europeans and European-Americans), so therefore they probably also contribute to the differences among ethnic groups. However, this conclusion is unfounded. To illustrate: Imagine that someone finds that the children in a certain city differ in their IQ scores and that part of this difference is due to genetics. Now, we take a random sample of the families and move them to a second city, with worse health, nutrition, education, and overall opportunities. We would probably find that the children now perform worse on the tests. If we compared children within one city or the other, *those* differences might be partly genetic, but the difference between cities could still be entirely due to environment. (See Figure 9.15.)

People who argue that heredity could not contribute to ethnic differences in IQ argue as follows: The various human ethnic groups have diverged for only a brief period of evolutionary history. The great majority of human genes either show no variation at all among individuals or differ widely *within*

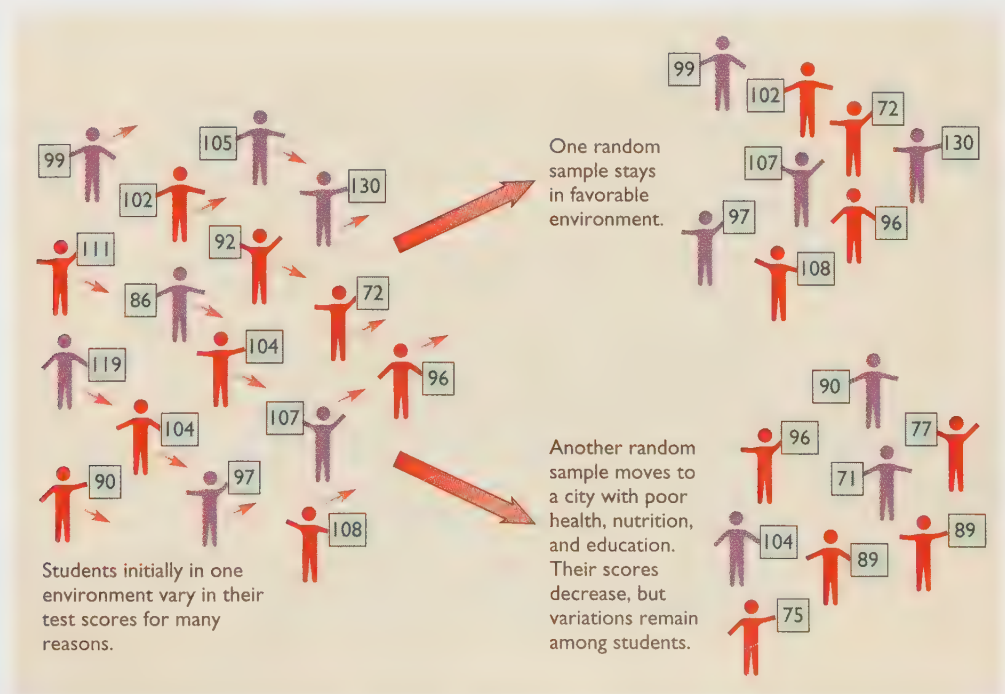
each ethnic group. Only a few genes (such as the genes controlling hair and skin color) have a greater difference among ethnic groups than they have within each group. (Actually, there are probably *no* genes that are present in all members of one group but none of another. At most, a gene might be present in a higher percentage of one group than the other.)

Granting all of that, the implications for the genetics of intelligence are still unclear, however. Different ethnic groups do share most of their genes in common, but within a given country such as the United States, different ethnic groups also share a good deal of their environment in common (Rowe, Vazsonyi, & Flannery, 1994). Furthermore, researchers do not know which genes, or even how many genes, make significant contributions to intelligence or how such genes exert their effects. Do such genes differ more within each human population than they differ among populations? Or are these genes some of the few that differ significantly between one population and another? In short, we do not yet know enough about either the environmental or the genetic influences on IQ to say what is possible. Arguing about probabilities and likelihoods is no substitute for collecting evidence.

Environmental Contributors to Ethnic Differences

Many studies attempting to evaluate the role of environmental factors have followed the design of providing improved early education, improved nutrition, or a change in

FIGURE 9.15 If two sets of children grow up in very different circumstances, it is possible that the difference between the two groups is entirely due to environmental differences, even if the differences within each group are partly due to genetic factors.





If ethnic differences in IQ are largely the result of environmental differences, African-American children adopted by upper-middle-class European-American families should score higher, on average, than similar children reared in African-American homes. The results of research have been unimpressive and also hard to interpret, because many of the children studied were adopted relatively late.

another environmental factor for an experimental group of extremely low-income children and then examining the effects on the children's eventual IQ scores and school performances. Most such studies have found genuine but fairly small benefits (e.g., Darlington, Royce, Snipper, Murray, & Lazar, 1980).

A more drastic approach is to examine African-American children who have been adopted by upper-middle-class

European-American families. The idea behind this approach is that such families are likely to provide a variety of environmental supports helpful for performance on IQ tests, including good nutrition, better education, and general familiarity with "the culture of the test."

Hypothesis If the IQ differences among the races is partly due to early experiences, then African-American children reared by European-American families should perform better on IQ tests than most other African-American children will.

Method Sandra Scarr and Richard Weinberg (1976) studied European-American families in Minnesota that had adopted African-American children (with two African-American parents) or ethnically mixed children (one African-American and one European-American parent). Many of these parents also had biological children of their own; many had also adopted European-American or Asian or Native American children. The investigators tested the IQs of the children and the adopting parents and tested them again 10 years later (Scarr & Weinberg, 1976; Weinberg, Scarr, & Waldman, 1992).

Results Figure 9.16 shows the mean scores for only those individuals who were tested both times. All the means were lower at the second test because of changes in the IQ test itself. (As mentioned earlier, psychologists restandardize the test every few years, generally making it more difficult.)

Interpretation The data of interest are the mean scores for the adopted children. The adopted European-American children scored higher than adopted children with one African-American parent ("ethnically mixed"), who in turn scored higher than adopted children with two African-American parents. However, the ethnically mixed adopted children scored near 100 (above 100 on the first test but

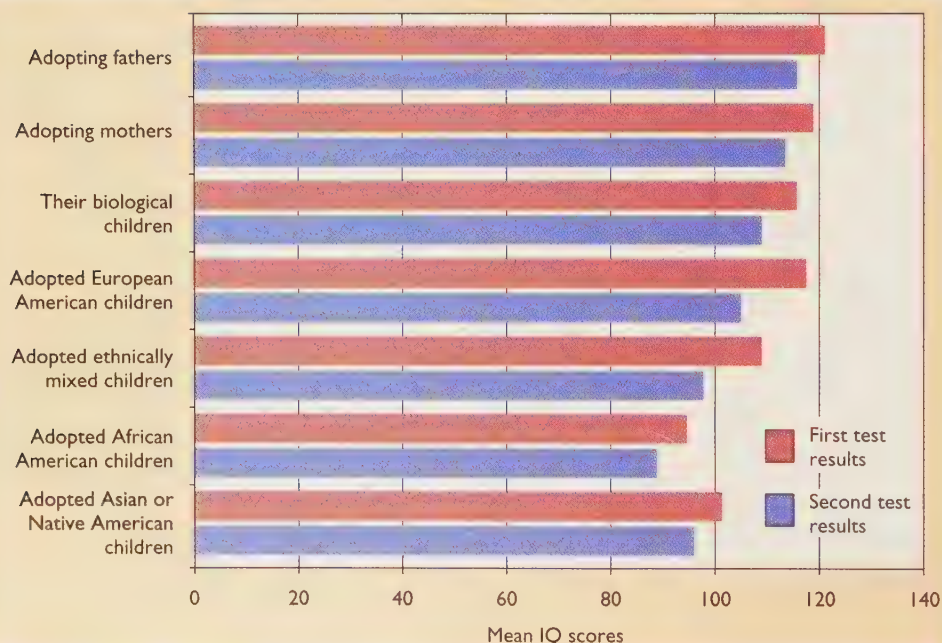


FIGURE 9.16 During childhood, African-American children adopted by upper-middle-class European-American families showed IQ scores well above the national average for African-Americans, but 10 years later they showed only small benefits.

slightly below that on the second). Adopted children with two African-American parents scored a mean of 95 on the first test; 89, ten years later.

Psychologists are frankly divided about how to interpret these results (e.g., Levin, 1994; Waldman, Weinberg, & Scarr, 1994). African-American children adopted by upper-middle-class European-American families did perform better on IQ tests than the national average for African-Americans (85), but not much better. Furthermore, the mean for African-Americans in Minnesota is not 85, but 88.

If these data suggest a disappointingly small environmental influence, one possible reason is that the presumed benefits of an “advantaged” family began well after birth. For whatever reason, the African-American mothers of the adopted children in this study gave their babies up for adoption significantly later than did the European-American mothers. Many of the adopted African-American and ethnically mixed children lived in several foster homes of unknown quality before being adopted. Perhaps these children demonstrated the effects of their very early environment, as well as their prenatal environment. (Many African-American mothers who give their babies up for adoption are teenagers who received poor health care and nutrition during pregnancy.) In short, this study, which was designed to demonstrate environmental influences, instead produced small and equivocal effects.



Hereditary Contributors to Ethnic Differences

In comparison, let us now examine a study with the potential to show a strong hereditary contribution.

Hypothesis If heredity is responsible for even part of the difference between the races in IQ performance, then African-Americans with a high percentage of European ancestry should obtain higher IQ scores than African-Americans with a lower percentage. (Few U.S. African-Americans have 100% African ancestry.)

Method Determining the ethnic ancestry of an African-American child is difficult. Most family trees do not go back enough generations, and skin color is an inaccurate indicator. (Even in Africa, skin darkness varies greatly from one subpopulation to another.) Blood-typing, however, provides a somewhat better estimator.

The investigators (Scarr, Pakstis, Katz, & Barker, 1977) examined 362 African-American children in Philadelphia, testing 14 different blood factors—the familiar ABO blood types, the Rh factor, the Duffy factor, and 11 others. Certain blood factors are more common in Europe than in Africa and vice versa. For example, type B blood is present in only 9% of Europeans but in 21% of Africans. Few if any



Variations in skin darkness are an inaccurate indicator of degree of African ancestry. Blood-typing provides a somewhat better indicator, although it is also far from certain.

Europeans have Duffy type A—B— blood, whereas 94% of Africans do. By comparing each child's blood factors to the frequency of those blood factors in both Europe and Africa, the investigators estimated the degree of European ancestry for each child. Then, the investigators correlated their estimates of European ancestry with performance on Raven's Progressive Matrices and four other tests of intellectual performance.

Results The investigators tried several methods of weighting the importance of various blood factors to estimate European and African ancestry. Regardless of which method they used, however, they found virtually zero correlation between the estimates of European ancestry and measures of performance.

Interpretation If hereditary factors were a major contributor to ethnic differences in IQ scores, we would expect to find a positive correlation between the amount of European ancestry and IQ scores. The absence of such a correlation indicates either that heredity contributes nothing to the ethnic differences or that it contributes so little that this research method could not detect it. Another possibility is that the investigators estimated African and European ancestry too inaccurately (Reed, 1997). Perhaps the main point here is simply that it is difficult to determine the roles of heredity and the environment for differences among human groups.



THE MESSAGE Heredity and Environment

Where does this leave us? The best designed studies, attempting to demonstrate an environmental contributor to ethnic differences, found only weak and uncertain effects. The best study capable of demonstrating a hereditary contributor found no evidence of any influence at all. Sometimes, even good research fails to answer the question

being posed; here, we have such a case. Future research may help to resolve the issues; in the meantime, the best recommendation is to draw no conclusion at all and to be suspicious of anyone who seems too sure of the answer.

What kind of research should psychologists do in the future? Actually, I would not encourage a young psychologist to begin research on the roles that heredity and environment play in IQ differences. Never mind the political explosiveness of the issue; the topic is scientifically difficult, and it is not clear what use (if any) the results would have. There is, however, one related topic that definitely does deserve more and improved research: Given that environmental factors do contribute to differences in intelligence (as everyone agrees, even if we don't all agree how much they contribute), what are the important environmental factors? Simply growing up in an upper-middle-class home is apparently insufficient to boost IQ or school performance by much. But what is sufficient? prenatal health and nutrition, perhaps? other factors in an infant's environment? If we could identify those factors, we might be able to accomplish something that would benefit everyone.

SUMMARY

- ★ **Standardization.** To determine the meaning of a test's scores, the authors of a test determine the mean and the distribution of scores for a random or representative sample of the population. IQ tests are revised periodically. To keep the same mean, test authors make the tests more difficult from time to time. (page 327)
- ★ **Distribution of IQ scores.** IQ tests have a mean of 100 and a standard deviation of about 15 or 16, depending on the test. Items are carefully selected so that performance on each item correlates positively with performance on the test as a whole. (page 327)
- ★ **Reliability and validity.** Tests are evaluated in terms of reliability and validity. Reliability is a measure of a test's consistency, or the repeatability of its scores. Validity is a determination of how well a test measures what it claims to measure. (page 328)
- ★ **Test bias.** Psychologists try to remove from a test any item that tends to be easy for one group of people to answer but difficult for another. They also try to evaluate the possible bias of a test as a whole. Bias is defined as a systematically incorrect estimation of how well a group will perform. By that definition, IQ tests are biased against immigrants but apparently not against racial minorities; they predict the school performance of African-Americans about as accurately as that of European-Americans. (page 332)
- ★ **Hereditary and environmental influences.** To determine the contribution of heredity to the variations in scores on IQ tests, investigators compare identical twins and fraternal twins, study identical twins reared apart, and compare adopted children with their biological and adoptive parents. For European-American families (about whom we have the most information), both hereditary and environ-

mental factors appear to contribute to observed differences in performance. (page 334)

- ★ **The controversy concerning ethnic differences in IQ.** Ethnic groups differ, on the average, in IQ performance, although the group mean scores cannot tell us what to expect from each individual. Research designed to show environmental contributors to these ethnic differences have generally produced disappointingly small effects that are difficult to interpret. The best-designed study capable of demonstrating a hereditary contribution failed to show even marginal effects. At present, the existing research evidence does not justify a confident conclusion about the origin of ethnic differences. (page 337)

Suggestion for Further Reading

Neisser, U., et al. (1996). Intelligence: knowns and unknowns. *American Psychologist*, 51, 77–101. Although much dispute surrounds the meaning of IQ scores and the reasons why individuals differ, psychologists have also found areas of agreement. This article describes the consensus of most psychologists today, the evidence behind their conclusions, and the questions that still remain unanswered.

Terms

- standardization** the process of establishing rules for administering a test and for interpreting the scores (page 327)
- norms** descriptions of the frequencies of occurrence of particular scores (page 327)
- Flynn effect** the tendency for people's performance on IQ tests to improve from one decade or generation to the next (page 328)
- reliability** the repeatability of a test's scores (page 328)
- test-retest reliability** the correlation between scores on a first test and on a retest (page 328)
- validity** the determination of how well a test measures what it claims to measure (page 329)
- content validity** the similarity between the items in a test and the information that the test is meant to measure (page 329)
- construct validity** the correspondence of a test's measurements to a theoretical construct (page 329)
- predictive validity** the ability of a test's scores to predict some real-world performance (page 329)
- utility** the usefulness of a test for a practical purpose (page 330)
- bias** the tendency for test scores to exaggerate a difference between groups or to report a nonexistent difference (page 333)

Answers to Concept Checks

3. Yes! To say that a test is "reliable" is simply to say that its scores are repeatable—that and only that. My test would give perfectly reliable (repeatable) measurements. True, they would be utterly useless, but that is beside the point. Reliability is not a measure of usefulness. (page 329)
4. No. An individual's score may be higher on the retest, either because of the practice at taking the test or because of the

additional months of education. But the rank order of scores does not change much. That is, if some people retake the test, all of them are likely to improve their scores, but those who had the highest scores the first time will probably also have the highest scores the second time. (page 329)

5. Yes, a test can have high reliability and low validity. A measure of intelligence determined by dividing head length by head width has high reliability (repeatability) but presumably no validity. A test with low reliability cannot have high validity, however. Low reliability means that the scores fluctuate randomly. If the test scores cannot even predict a later score on the same test, then they can hardly predict anything else. (page 330)
6. The predictive validity of the tests would increase. The predictive validity tends to be low when almost all students have practically the same score; it is higher when students' scores are highly variable. (page 330)
7. The predictive validity of SAT scores will be higher at the university that admits almost anyone. At the university with extremely competitive admissions standards, almost all students have nearly the same SAT scores, and the slight variation in scores cannot predict which students will get the best grades. (page 330)

8. No, this test is not biased against people with visual impairments. It correctly determines that they are likely to be poor drivers. (page 334)
9. We would need to determine whether the test accurately predicts the school performances of both short and tall people. If short people with, say, an IQ score of 100 perform better in school than tall people with an IQ score of 100, then the test is underpredicting the performances of short people, and we can thus conclude that this test is biased against them. (Merely that the test reports a difference between short people and tall people is not in itself evidence of test bias.) (page 334)

Web Resources

The Bell Curve and Its Critics

www.cycad.com/cgi-bin/Upstream/People/Murray/bc-crit.html
Charles Murray, co-author of *The Bell Curve*, responds to criticism of the ideas advanced in the book.

Development



10

★ MODULE 10.1

The Study of Early Development

The Fetus and the Newborn

Prenatal Development

Behavioral Capacities of the Newborn

Newborns' Vision / Newborns' Hearing / The Learning and Memory of Newborns / INFANTS' THOUGHT

PROCESSES ABOUT OBJECT PERMANENCE: WHAT'S THE EVIDENCE? / Further Capacities of the Infant

★ MODULE 10.2

The Development of Thinking and Reasoning

The Development of Thought and Knowledge:

Piaget's Contributions

Intellectual Growth: Some Piagetian Terminology /

Piaget's Stages of Intellectual Development / The

Sensorimotor Stage: Infancy / The Preoperational Stage:

Early Childhood / Egocentric Thinking in the

Preoperational Period / CHILDREN'S UNDERSTANDING OF

OTHER PEOPLE'S COGNITIONS: WHAT'S THE EVIDENCE? /

Lack of the Concept of Conservation in the

Preoperational Period / The Concrete Operations Stage:

Later Childhood / The Formal Operations Stage:

Adolescence and Adulthood / Are Piaget's Stages

Distinct? / Implications for Education: Piaget and Lev

Vygotsky

The Development of Moral Reasoning

Kohlberg's Method of Evaluating Levels of Moral

Reasoning / The Limitations of Kohlberg's Views of

Moral Development

★ MODULE 10.3

Social and Emotional Development

Research Designs for Studying Development

Erikson's Ages of Human Development

Infancy: Forming the First Attachments

Studies of Attachment Among Monkeys / Early

Attachment in Humans / The Effects of Different

Parenting Styles

Social Development in Childhood

Adolescence

Identity Development / The "Personal Fable" of

Teenagers / Job Satisfaction / The Midlife Transition

Adulthood and Aging

Old Age

The Psychology of Facing Death

★ MODULE 10.4

Growing Up the Same and Different: Temperament, Family, Gender, and Cultural Influences

Temperament and Lifelong Development

The Family

Parental Employment and Child Care / Nontraditional

Families / Parental Conflict and Divorce

The Influence of Gender

Sex Differences in Behavior

Ethnic and Cultural Influences

Suppose you buy a robot. When you get home, you discover that it does nothing useful. It cannot even maintain its balance. It makes irritating, high-pitched noises, moves its limbs about haphazardly, and leaks. The store you bought it from refuses to take it back. And you discover that, for some reason, it is illegal to turn it off. So you are stuck with this useless machine.

A few years later, your robot can walk and talk, read and write, draw pictures, and do arithmetic. It will follow your directions (most of the time), and sometimes it will even find useful things to do without being told. It beats you consistently at checkers and destroys you at memory games.

How did all this happen? After all, you knew nothing about how to program a robot. Did your robot have some sort of built-in programming that simply took a long time to phase in? Or was it programmed to learn all these skills by imitating what it saw?

Children are a great deal like that robot. Nearly every parent wonders, "How did my children get to be the way they are?" The goal of developmental psychology is to understand everything that influences human behavior "from womb to tomb."



As we grow older, we change in many ways—we gain in some ways and lose in others. Developmental psychologists seek to understand the changes in our behavior and the reasons behind these changes.

The Study of Early Development

What are the capacities of the newborn and the young infant?

How can psychologists determine those capacities?

The artwork of young children can be amazingly inventive and can reveal a great deal about what the children are thinking. One toddler, 1½ years old, showed off a drawing that consisted only of dots on a sheet of paper. Puzzled adults did not understand the drawing. It is a rabbit, the child explained, while making more dots: “Look: hop, hop, hop. . .” (Winner, 1986).

When my daughter, Robin, was 6 years old, she drew a picture of a boy and a girl drawing pictures (Figure 10.1). The overall drawing has features that may not be clear; for example, both children are wearing Halloween costumes.

For the little girl’s drawing, Robin pasted on some wildlife photos. This array, she maintained, was what the little girl had drawn. Now look at the little boy’s drawing: It’s just a scribble. When I asked why the little girl’s drawing was so much better than the little boy’s, Robin replied, “Don’t make fun of him, Daddy. He’s doing the best he can.”

Sometimes, as in this case, a child’s drawing can tell us a great deal about the child’s worldview. As children grow older, their art changes. Robin Kalat, now a high-school student, draws pictures with a skill that I envy. Still, I sometimes miss the highly expressive drawings of her early childhood.

The point is this: As we grow older, we develop; we gain many new abilities and skills. But we lose something too.

Studying the abilities of children, especially infants and very young children, is extremely challenging. The very young are often capable of far more than we realize, simply because they misunderstand our questions or we misunderstand their answers, or because we assume that, if they cannot do very much, they probably don’t think very much either. It is also easy to underestimate the abilities of the very old. Developmental psychologists have made much progress by devising increasingly careful and sensitive ways to measure behavioral abilities.

The Fetus and the Newborn

Even at the time of birth, a human infant can see, hear, and show indications of a certain amount of learning and memory. Within the next few weeks or months, the infant is ready to acquire new skills at an impressive rate.



FIGURE 10.1 A drawing of two children drawing pictures, courtesy of 6-year-old Robin Kalat.

Prenatal Development

During prenatal development—development before birth—everyone starts life as a fertilized egg cell, or zygote. That fertilized egg develops through several stages, known as blastula, gastrula, embryo (from about 2 to 8 weeks after conception in humans), and fetus (from about 8 weeks after conception until birth, in humans). A human embryo does not look much different from a hog embryo or that of any other vertebrate (Figure 10.2).

Even a human fetus still has a long way to go. All of its organs, including the brain, must mature a great deal before birth. Different structures and substructures mature at different times; consequently, traumas and teratogens (poisons) produce various kinds of impairments at different ages.

The growing body receives its nutrition from the mother. If she eats little, the baby receives little nourishment. If she takes drugs, the baby gets them too. Undernourished mothers generally give birth to small babies (Figure 10.3), and investigators have long known that newborns



FIGURE 10.3 Advances in modern medicine enable us to keep babies alive even when they are born premature or very small. Low-birth-weight babies are prone to a variety of physical and behavioral difficulties later in life; however, we cannot be sure that low birth weight causes these problems. Many of these babies are born to mothers who are very young or impoverished, who take illegal drugs, or who fail to provide good nutrition and care.



FIGURE 10.2 During the earliest stages of development, a human embryo looks much the same as the embryo of any other vertebrate species. The differences emerge later.

weighing less than about 1,750–2,000 grams (4 pounds) are at high risk of dying in infancy (Kopp, 1990). If such babies survive, they have an increased risk of eventual mental retardation, low academic achievement, and various behavior problems (Morgan & Winick, 1989). Those facts are beyond dispute, but let's consider how difficult it is to interpret them.

The apparently obvious interpretation is that low birth weight leads to impaired brain development and thus to later academic and behavior problems. However, many low-birth-weight babies are born to mothers who are poor, uneducated, unhealthy, teenaged, unmarried or unhappily married, victims of family violence, having a low-vitamin or low-mineral diet, possibly smoking or drinking during pregnancy, not visiting a doctor for care and advice during pregnancy, and unable to provide a good home after delivering the baby (Garcia Coll, 1990; McCormick, 1985). In short, many low-birth-weight babies have other, probably more serious problems as well, which combine to increase the risk to the child (Brooks-Gunn & Furstenberg, 1989; Zeanah, Boris, & Larrieu, 1997).

One way to study the effect of low birth weight separately from other problems is to examine pairs of twins where one twin was born much heavier than the other. The twins differ in birth weight but have the same parents and the same environment both before and after birth. In most cases, the low-birth-weight infant develops about as well as the heavier twin (Wilson, 1987). In short, low birth weight by itself is not necessarily a serious problem; it correlates with developmental difficulties partly because many low-birth-weight babies have other disadvantages.

A more severe risk arises if the fetus is exposed to alcohol or other substances. If the mother drinks alcohol during pregnancy, the infant may develop signs of fetal alcohol

syndrome, a condition marked by stunted growth of the head and body; malformations of the face, heart, and ears; and nervous system damage, including seizures, hyperactivity, learning disabilities, and mental retardation (Streissguth, Sampson, & Barr, 1989). In milder cases, the child's appearance may be normal, but the child is still likely to have impaired academic skills and deficits in language, memory, and fine motor speed (Mattson, Riley, Grambling, Delis, & Jones, 1998). The more alcohol the mother drinks and the longer she drinks during pregnancy, the greater the risk to the fetus (see Figure 10.4). Similarly, women who smoke during pregnancy have an increased probability that their babies will be born very small and that the babies will have a variety of serious health problems early in life. These mothers also run an increased risk that their children, especially sons, will develop conduct disorder, a condition marked by discipline problems both at school and at home, and potentially even criminal behavior in adulthood. Conduct disorder has been found to correlate more strongly with the mother's smoking during pregnancy than with the father's antisocial behavior, the family's economic status, lack of supervision of the child, or excessive punishment (Wakschlag et al., 1997). To be safe, pregnant women should avoid alcohol and tobacco and should get a physician's advice before taking even routine, over-the-counter drugs.

Still, it is remarkable that an occasional "high risk" child—small at birth, perhaps exposed to alcohol or other drugs before birth, perhaps from an impoverished or turbulent family—overcomes all odds and becomes a healthy, productive, outstanding person (Werner, 1989). Resilience (the ability to overcome obstacles) is hardly well understood, but the general pattern is that the more positive influences one has in life, the easier it is to overcome a negative one. That is, people who overcome a great disadvantage tend to have some special source of strength—a close relationship with one or more supporting people, an effective school, a strong faith, some special skill, or just a naturally easygoing disposition (Masten & Coatsworth, 1998).

Behavioral Capacities of the Newborn

A human newborn is a little like a computer that is not attached to a monitor: It may be processing a great deal of information, but it cannot tell us about it. The challenge of studying the newborn is to figure out how to attach some sort of "monitor" to find out what is going on inside the newborn's head.

Newborns have very little control of their muscles. At first, they cannot keep their head from flopping over, and their arms and legs flail about aimlessly. They gradually gain more muscle control, partly through growth and maturation of the muscles and nerves and partly from practice. For example, babies are much more persistent at waving their arms if they can see their arms than if they have to wave without watching (van der Meer, van der Weel, & Lee, 1995). Also, babies who are too young to support their own weight flail their arms and legs in varied and haphazard ways; as soon as they are capable of supporting their weight, they quickly abandon most of their varied movement patterns and settle on the standard patterns and rhythms of human crawling (Freedland & Bertenthal, 1994).

About the only useful movements that newborns can make are mouth and eye movements. As the months pass, and as their control spreads from the head muscles downward, they are able to make progressively finer movements, eventually culminating in the ability to move a single finger at a time.

If we want to test an infant's sensory and learning abilities, we must test them by means of responses that the infant can control. For example, if we want to test what an infant can see, we should examine eye movements or head movements; if we try to train an infant to reach out and grab something, we will almost certainly underestimate the infant's capacities.

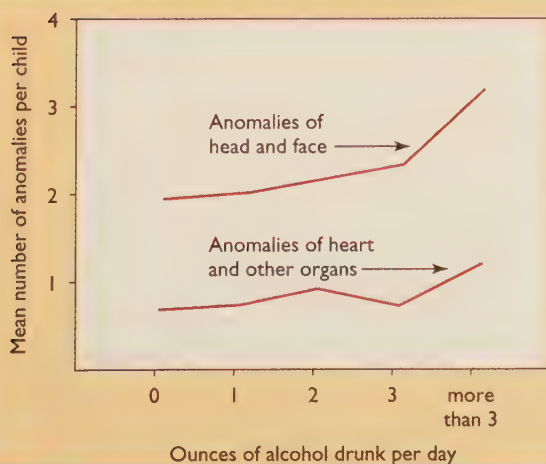


FIGURE 10.4 (a) The more alcohol a woman drinks during pregnancy, the more likely her baby is to have anomalies of the head, face, and organs. (Based on data of Ernhart et al., 1987.) (b) A child with fetal alcohol syndrome: Note especially the wide separation between the eyes, a common feature of this syndrome.

Newborns' Vision

William James, a pioneer in American psychology, once said that as far as an infant can tell, the world is a “buzzing confusion,” full of meaningless sights and sounds. Since James’s time, psychologists have substantially increased their estimate of what an infant can see.

One research method is to record the infant’s eye movements. In general, infants direct their eyes toward the same kinds of objects that attract adults’ attention. For example, even 2-day-old infants spend more time looking at drawings of human faces than at other patterns with similar areas of light and dark (Fantz, 1963). (See Figure 10.5.)

By the age of 5 months or so, infants have had extensive visual experience but almost no experience at crawling or reaching for objects. As they start to gain control of their arm and leg movements, they suddenly have to reach out to pick up toys, crawl around objects, avoid crawling off ledges, and in other ways coordinate what they see with what they do.

Apparently, infants must have some experience of controlling their own movements before they show a fear of heights. Infants begin to show a fear of heights shortly after they begin to crawl—presumably also the age when they have their first experience of slipping and falling. Infants who crawl early develop a fear of heights early; infants who are late to crawl are also late to develop a fear of heights (Campos, Bertenthal, & Kermoian, 1992).

Although visual-motor coordination develops quickly, infants need practice to maintain and improve that coordination. Kittens are ideal for studies of visual-motor development because they can move about quite well by the time they first open their eyes. In one experiment, kittens were permitted to walk around in a dark room for 21 hours a day (Held & Hein, 1963). For the other 3 hours, half of the kittens (the “active” group) were permitted to walk around in a well-lit cylindrical room, as Figure 10.6 shows. The other kittens (the “passive” group) were confined to boxes that were propelled around the room by the active kittens.

The active kittens gradually developed good paw-eye coordination, but the passive kittens lagged far behind. In

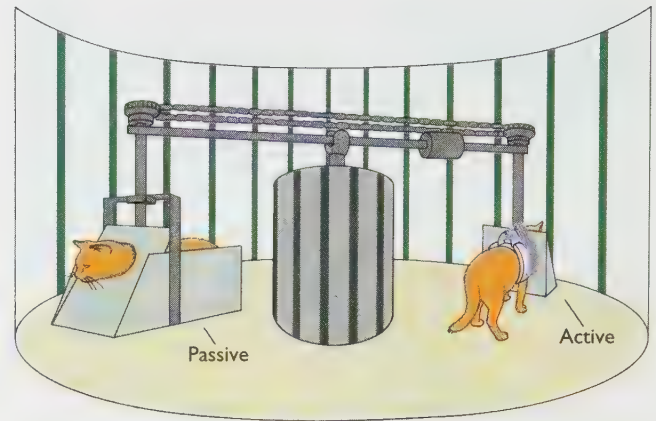


FIGURE 10.6 As the kitten carousel experiment demonstrates, experience influences development. These two kittens see the same thing, but only one can correlate what it sees with its own movements. Only the active kitten develops normal paw-eye coordination. (Modified from Held & Hein, 1963.)

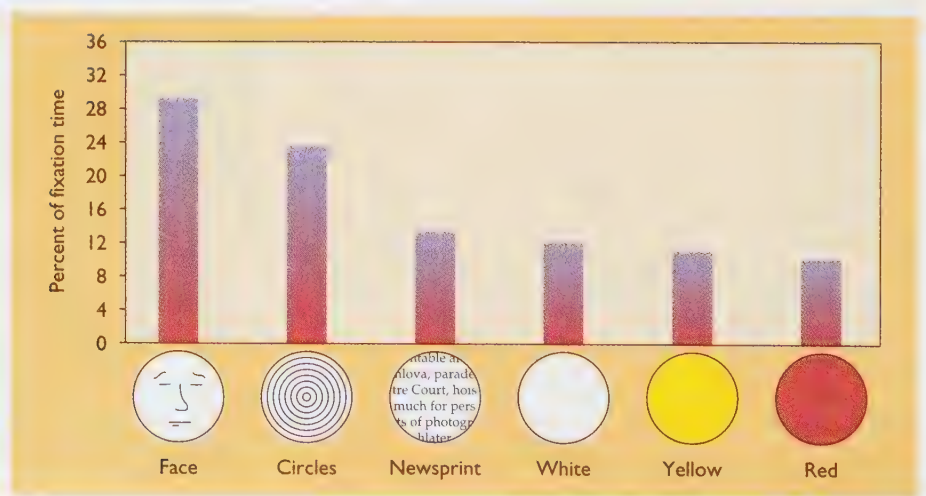
fact, the passive kittens’ coordination actually grew worse instead of better as the experiment continued. Evidently, kittens need to see and move at the same time in order to maintain and improve their visually guided behavior. The same is almost certainly true for humans as well.

Newborns' Hearing

At first it might seem difficult to measure newborns’ responses to sounds; after all, we cannot observe anything similar to eye movements. However, we can record the effects of sounds on an infant’s sucking. Infants suck more vigorously when they are aroused, and certain sounds arouse them more than others do.

In one study, the experimenters played a brief sound and noted how it affected the infant’s sucking rate (Figure 10.7). On the first few occasions, the sound increased the sucking rate. After the sound had been played repeatedly, it produced less and less effect. We say that the infant became habituated to the sound. Habituation is decreased response

FIGURE 10.5 Infants pay more attention to faces than to other patterns. (Based on Fantz, 1963.) These results suggest that infants are born with certain visual preferences that do not depend on learned associations. Their preference for faces facilitates the development of social attachments.



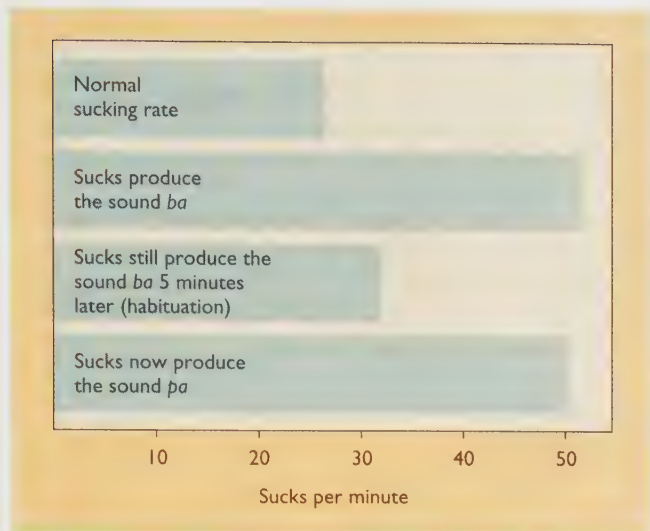


FIGURE 10.7 After 5 minutes of hearing the same sound, the infant's sucking habituates. When a new sound, *pa*, follows, the sucking rate increases, an indication that infants do hear a difference between the two sounds *ba* and *pa*. (Based on results of Eimas, Siqueland, Jusczyk, & Vigorito, 1971.)

to a repeated stimulus. When the experimenters substituted a new sound, it produced a sharp increase in the sucking rate. Evidently, the infant was aroused because he or she heard a new, unfamiliar sound. When a change in a stimulus produces an increase in a previously habituated response, we say that the stimulus has produced dishabituation.

Psychologists use this technique to determine whether infants hear a difference between two sounds (Jusczyk, 1985). For example, an infant who has become habituated to the sound *ba* will increase the sucking rate in response to the sound *pa* (Eimas, Siqueland, Jusczyk, & Vigorito, 1971). Apparently, even month-old infants can tell the difference between *ba* and *pa*. Evidently, infants have some preliminary skills of attending to language sounds long before they develop other language skills.

The Learning and Memory of Newborns

Infants certainly cannot describe their memories to us. But if they respond differently to a stimulus because of previous experience with it, we can infer that they remember it.

Several studies have begun with the observation that infants learn to suck harder on a nipple if their sucking turns on a sound. Investigators then tried to determine whether infants will work harder to turn on certain sounds than they will for others. In one study, 26 babies less than 3 days old could turn on a tape recording of their mother's voice by sucking on a nipple at certain times and at certain rates. By sucking at different times or at different rates, they could turn on a tape recording of another woman's voice. When their manner of sucking produced their own mother's voice, their rate of sucking increased significantly (DeCasper & Fifer, 1980); it increased less when it produced a different voice. Apparently, even very young infants recog-



FIGURE 10.8 Inspired by research showing that a fetus learns to recognize its mother's voice, some women have made special efforts to talk to their fetuses—and some enterprising capitalists have sold them “pregaphones,” manuals, tapes, and lessons. (Don't assume that the fetus is the one getting the greatest benefit from all of this.)

nized their own mother's voice and preferred it to an unfamiliar voice. Because they showed this preference so early—in some cases, on the day of their birth—developmental psychologists believe that the infants are displaying a memory of what they heard before birth (Figure 10.8).

CONCEPT CHECK

- I. Suppose that a newborn sucks to turn on a tape recording of its father's voice. Eventually, the baby habituates and the sucking rate decreases. Now, the experimenters substitute the recording of a different man's voice for the father's. What would you conclude if the sucking rate increased? What if it remained the same? What if it decreased? (Check your answers on page 353.)

Using somewhat older infants, Carolyn Rovee-Collier (1984) demonstrated an ability to learn a response and remember it for days afterward. She attached a ribbon to one ankle so that an infant could activate a mobile by kicking with one leg (Figure 10.9); 2-month-old infants quickly learned this response and generally kept the mobile going nonstop for a full 45-minute session. (I know I said infants cannot control their leg muscles, but they don't need much control to keep the mobile going.) Once they have learned, they quickly remember what to do when the ribbon is reattached several days later—to the infants' evident delight.

WHAT IS THE EVIDENCE? The Infant's Thought Processes About Object Permanence

Measuring an infant's vision, hearing, and memory can be difficult enough. Inferring the infant's thoughts and knowledge is trickier yet, although researchers have found some clever



FIGURE 10.9 By the age of 8 weeks, infants can rapidly learn to kick one of their legs to activate a mobile attached to their ankle with a ribbon. After just a little practice, they can keep the mobile going for a full 45-minute session. These infants also remember how to activate the mobile from one session to the next. (From Rovee-Collier, 1984.)

ways to do just that. As you can imagine, however, these inferences must be made cautiously and tentatively.

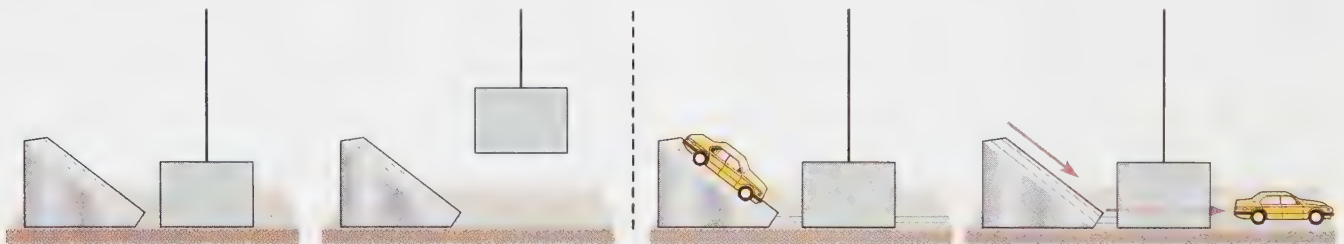
Jean Piaget (peah-ZHAY), whose theories we shall consider in more detail in the next module, inferred that infants in the first few months of life lack the concept of object per-

toy in front of a 6-month-old infant, who reaches out and grabs it. Later, place a toy in the same place, but before the infant has a chance to grab it, cover it with a clear glass. No problem; the infant removes the glass and takes the toy. Now, you repeat that procedure, but this time you cover the toy with an opaque (nonclear) glass. The infant, who watched you place the glass over the toy, makes no effort to remove the glass and obtain the toy. Or you put the toy down and then place a thin barrier between the infant and the toy. The infant will reach for a toy that is partly visible but not for one that is completely hidden (Piaget, 1937/1954). (See Figure 10.10.)

Why not? According to Piaget, the infant *does not know* that the hidden toy is there; not until about the age of 9 months does he or she fully understand object permanence. However, one can imagine other possible explanations for Piaget's results. One later study suggests that infants do show signs of understanding object permanence when they are tested differently.

Hypothesis An infant who sees an event that should be impossible (if objects are permanent) will stare longer than will an infant that sees a similar but possible event.

Method Infants aged 6 or 8 months watched a series of events staged by the researcher. First, the child watched the experimenter raise a screen to show nothing behind it and then watched a toy car go down a slope and emerge on the other side, as shown in the drawing below. The researchers measured how long the child stared after the car went down the slope; they repeated the procedure until the child decreased the staring time for three trials in a row (showing habituation).



manence, the idea that objects continue to exist even when we do not see or hear them. How could he know? He drew the inference from observations of the following type: Place a

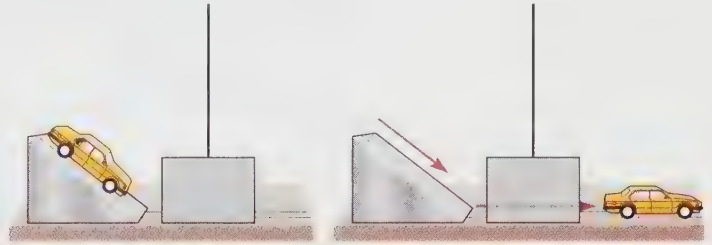
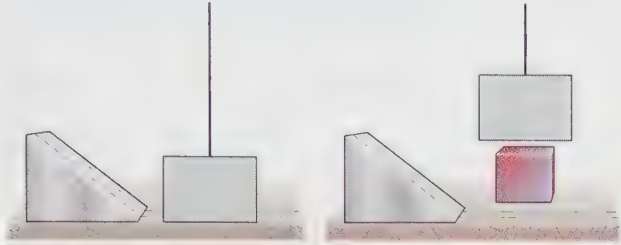
Then, the experimenters presented two kinds of events, three times each, as shown on page 351. In a Possible event, the raised screen showed a box that was behind the screen but not

FIGURE 10.10 During the sensorimotor period, a child will reach for a visible toy (left) but not one that is hidden behind a barrier (right)—even if the child sees someone hide the toy. According to Piaget, this observation indicates that the child lacks the concept of object permanence.

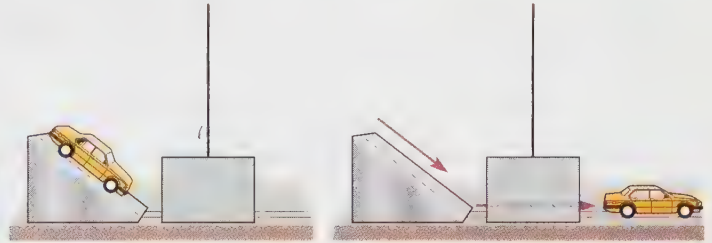
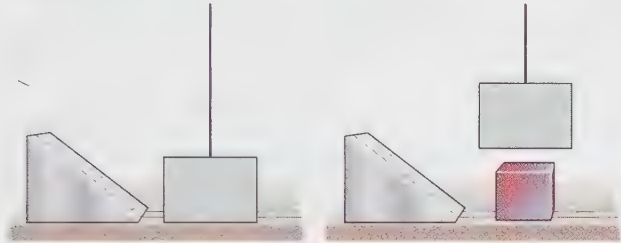


on the track. In an Impossible event, the raised screen showed a box that was on the track, right where the car would pass. After the screen lowered, the car went down the slope and (for both Possible and Impossible events) emerged on the other side. (For Impossible events, the experimenters had pulled the box off the track after lowering the screen.) The experimenters measured each child's staring times after both kinds of events.

Possible event



Impossible event



Results Figure 10.11 shows the mean looking times. Infants stared longer after seeing an Impossible event than after seeing a Possible event. They also stared longer during the first pair of events than after the second pair and longer after the second than the third (Baillargeon, 1986).

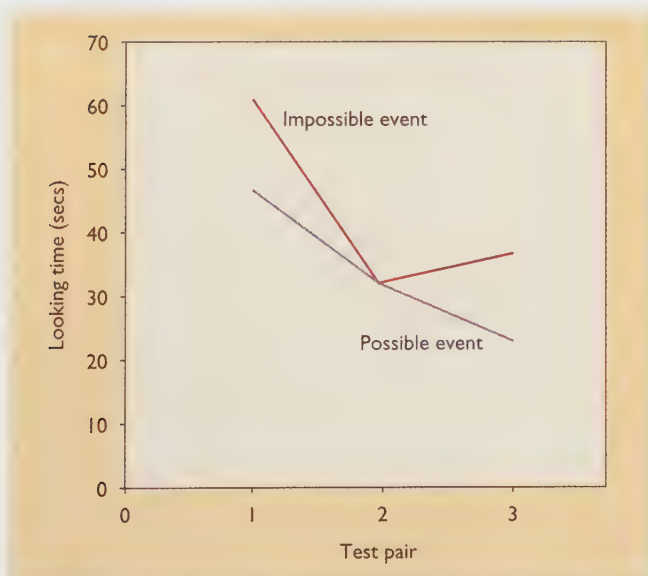


FIGURE 10.11 Mean looking times of 6- and 8-month-old infants after they had watched either Possible or Impossible events. (From Baillargeon, 1986.)

Interpretation Why did the infants stare longer at the “Impossible” event? The inference—and admittedly it is only an inference, not a certainty—is that the infants found the Impossible event surprising. To be surprised, the infants had to expect that the box would continue to exist where it was hidden and that a car could not go through it. If this inference is correct, even 6-month-old infants understand a

fair amount about the permanence of objects. One later study with a slightly different method again measured how long infants stared at Possible and Impossible events and demonstrated object permanence in infants as young as 3½ months (Baillargeon, 1987).

Still, remember that infants as old as 9 months failed Piaget’s object permanence task of reaching out to pick up a hidden object. Do infants have the concept of object permanence or do they lack it? Perhaps there is something wrong with that question. It is possible to “have” a concept but use it in some situations and not in others (Munakata, McClelland, Johnson, & Siegler, 1997). Even college students can pass a physics test and then fail to apply the laws of motion in a new situation or understand the rules of grammar and yet make grammatical errors.

The important point about research is this: Especially when we are dealing with infants or anyone else whose thought processes are likely to be different from our own, we should be cautious about inferring what they can or cannot do. The results may vary depending on exactly what procedures and tests we use.



Further Capacities of the Infant

Using the technique of measuring infants’ staring time, researchers have also made a surprising inference about infants’ concept of number: Karen Wynn (1992) showed an

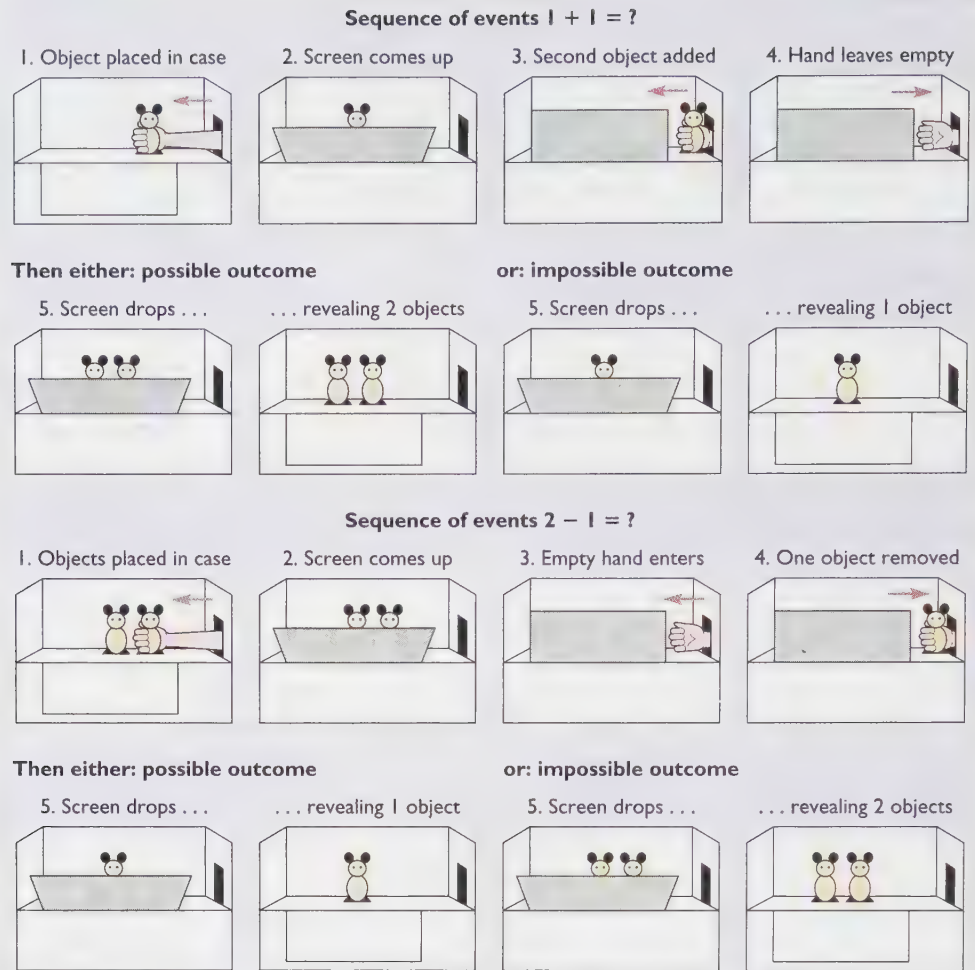


FIGURE 10.12 Five-month-old infants stare longer at an “impossible” display than at the “possible” display. Evidently, even at this age, children understand that adding or subtracting an object should change the number present.

infant an object, then raised a screen, and then placed another object behind the screen. Or else she showed the infant two objects, raised a screen, and then removed one of the objects from behind the screen. Finally, the screen dropped, revealing either the correct, expected number of objects ($1 + 1 = 2$ or $2 - 1 = 1$) or an “impossible” total ($1 + 1 = 1$ or $2 - 1 = 2$). Figure 10.12 shows these procedures. On the average, 5-month-old infants stared longer at an impossible total than at an expected total. Evidently, even at the age of 5 months, infants have a primitive concept that adding something should make the total more and subtracting should make it less.

poor control of their muscles, they are indeed like a computer with a faulty monitor: They have a lot more going on inside than they can show us.

SUMMARY

★ **Prenatal development.** Behavioral development begins before birth. During prenatal development, an individual is especially vulnerable to the damaging effects of alcohol and other drugs. Babies who are very small at birth have a high risk of later problems, mainly because they generally do not get good care or stimulation later in life. (page 346)

★ **Inferring infant capacities.** It is easy to underestimate the capacities of newborn human infants, because they have so little control over their muscles. With careful testing procedures, we can demonstrate that newborns can see, hear, and remember more than we might have supposed. (page 347)

★ **Inferring infant thought processes.** Infants behave differently from older children in many ways. For example, infants fail to reach for a toy after watching someone hide it within their reach. We can try to draw inferences about infants’ thinking, but we must be cautious about these inferences. An infant who fails to reach for a toy may not know

THE MESSAGE

*Infants—Capable of So Little
and So Much*

Notice that the experiments just described enable psychologists to explore infant thought and perception long before the infants can talk. The trend has been a steady increase in our estimation of infants’ capacities. Because of infants’

the toy still exists, may be distracted by other sights, may have forgotten the hiding place, or may be following a motor habit that is inconsistent with reaching in the correct direction. (page 349)

Suggestion for Further Reading

Mandler, J. M. (1990). A new perspective on cognitive development in infancy. *American Scientist*, 78, 236–243. An interesting article describing the often surprising capabilities of infants less than 6 months old.

Terms

prenatal development development before birth (page 346)

zygote a fertilized egg cell (page 346)

fetus an organism more developed than an embryo but not yet born (from about 8 weeks after conception until birth, in humans) (page 346)

fetal alcohol syndrome a condition marked by stunted growth of the head and body; malformations of the face, heart, and ears; and nervous system damage, including seizures, hyperactivity, learning disabilities, and mental retardation (page 346)

habituation a decrease in a person's response to a stimulus after it has been presented repeatedly (page 348)

dishabituation an increase in a previously habituated response as a result of a change in the stimulus (page 349)

object permanence the concept that objects continue to exist even when one does not see, hear, or otherwise sense them (page 350)

Answer to Concept Check

1. If the rate increased, we would conclude that the infant recognizes the difference between the father's voice and the other voice. If the rate remained the same, we would conclude that the infant did not notice a difference. If it decreased, we would assume that the infant preferred the sound of the father's voice for some reason. (page 349)

Web Resources

Classic Theories of Child Development

idealist.com/children/cdw.html

In a hypertext tutorial about the theories of Freud, Mahler, and Erikson, Linda Chapman also takes a month-by-month look at the normal developmental sequence from birth to three years.

The Development of Thinking and Reasoning

What goes on in the mind of a small child?

How do the thoughts of children differ from those of adults?

How does moral reasoning develop?

Preschool children ask some profound questions: “Why is the sky blue? What makes ice cubes cold? If it’s dangerous to look at the sun, why is it safe to look at a picture of the sun? Where does the sun go at night?” They are relentlessly curious about how things work and why. (Moreover, when you answer their questions, they never interrupt to ask, “Is this going to be on the test?”)

These same budding little scientists also believe in Santa Claus and the tooth fairy. Adults find it difficult to recapture what it is like to be a child. It is clear that children think differently from adults in a variety of ways, although it is not easy to specify those ways. Nevertheless, we try.

The Development of Thought and Knowledge: Piaget’s Contributions

Attending a rousing political rally can have a profound effect on a young adult, less effect on a preteen, and no effect at all on an infant. However, playing with a pile of blocks will be a more stimulating experience for a young child than for someone older. The effect of a certain experience on a person’s thinking processes and knowledge depends on the level of that person’s maturity and his or her previous experiences. The theorist who made this point most strongly and most influentially was Jean Piaget (1896–1980). (See Figure 10.13.)

Early in Piaget’s career, he administered IQ tests to French-speaking children in Switzerland. He grew bored with the IQ tests because he felt he was not learning anything about intelligence, but he was fascinated by the incorrect answers that children consistently gave in response to certain questions. For example, when asked, “If you mix

some water at a temperature of 50 degrees with an equal amount of water at 70 degrees, what temperature will the mixture be?” most 9-year-olds answer, “120 degrees” (Jensen, 1980).

Unless someone was going around mischievously misinforming all the children in Switzerland, the children must be reaching incorrect conclusions on their own. In other words—and this is one of Piaget’s central insights—children’s thought processes are different from those of adults. Children are not merely inexperienced adults, and they are not just less skillfully going through the same thought processes that adults use. The difference between children’s thought processes and those of adults is qualitative as well as quantitative—that is, it is both a difference in kind and a difference in amount. Piaget supported this conclusion with extensive studies of children, especially his own.

Intellectual Growth: Some Piagetian Terminology

According to Piaget, a child’s intellectual development is not merely an accumulation of experience or a maturational unfolding. Rather, the child constructs new mental processes as he or she interacts with the environment.

In Piaget’s terminology, behavior is based on schemata (the plural of *schema*). A schema is an organized way of interacting with objects in the world. For instance, infants



FIGURE 10.13 Jean Piaget (1896–1980), the most influential theorist on intellectual development in children, demonstrated that the influence of an experience on a person’s way of thinking depends on that person’s age and previous experience.



FIGURE 10.14 According to Piaget, assimilation and accommodation occur whenever we interact with an object. Here, an infant assimilates new objects to the grasp schema, applying an established behavior to them. However, the infant also accommodates the grasp schema, adjusting it to fit objects of different shapes and sizes.

have a grasping schema and a sucking schema. Older infants gradually add new schemata to their repertoire and adapt their old ones. This adaptation takes place through the processes of assimilation and accommodation.

With **assimilation**, a person *applies an old schema to new objects*—for example, an infant may suck an unfamiliar object or use the grasp response when trying to manipulate it. With **accommodation**, a person *modifies an old schema to fit a new object*—for example, an infant may suck a breast, a bottle, and a pacifier in different ways or modify the grasp response to accommodate the size or shape of a new toy (Figure 10.14).

Infants shift back and forth from assimilation to accommodation. For example, an infant who tries to suck on a rubber ball (assimilating it to her sucking schema) may find that she cannot fit it into her mouth. First, she may try to accommodate her sucking schema to fit the ball; if that fails, she may try to shake the ball. She is assimilating the new object to her grasping schema, but she is also accommodating that schema—changing it—to fit the ball.

Adults do much the same thing. You are given a new mathematical problem to solve. You try several of the methods you have already learned until you hit upon the one schema that works. In other words, you assimilate the new problem to your old schema. However, if the new problem is quite different from any problem you have ever solved before, you modify (accommodate) your schema until you work out a solution. Through processes like these, said Piaget, intellectual growth occurs.

Piaget's Stages of Intellectual Development

Piaget contended that children progress through four major stages of intellectual development:

1. The sensorimotor stage (from birth to about 1½ years)
2. The preoperational stage (from about 1½ to 7 years)
3. The concrete-operations stage (from about 7 to 11 years)
4. The formal-operations stage (from about 11 years onward)

The ages given here are only approximate, and Piaget was rightly more interested in the sequence of stages than the exact age at which someone reaches a particular stage

(Lourengo & Machado, 1996). Let us consider the capacities of children at each of these stages.

The Sensorimotor Stage: Infancy

Piaget called the first stage of intellectual development the **sensorimotor stage**, because at this early age (birth to 1½ years) behavior consists mostly of simple motor responses to sensory stimuli—for example, the grasp reflex and the sucking reflex. The fact that infants do not look for objects that they cannot see indicated to Piaget that infants respond only to what they see and hear, rather than to what they might remember or imagine.

Infants do, nevertheless, notice relationships among their experiences (Mandler, 1990). For example, even infants as young as 4 months old pay more attention to a film when the sound track matches the action than one in which the sound track is unrelated to the action. Also, an infant who has been sucking on a pacifier will stare longer at a pacifier of the same shape than a pacifier of a different shape.

As children progress through the sensorimotor stage of development, they appear to gain some concept of self. The data are as follows: A mother puts a spot of unscented rouge on an infant's nose and then places the infant in front of a mirror. Infants less than 1½ years old either ignore the red spot they see on the baby in the mirror or else reach out to touch the red spot. At some point after age 1½ years, infants in the same situation will touch themselves on the nose, indicating that they recognize themselves in the mirror (Figure 10.15). Infants show this sign of self-recognition at varying ages; the age when they start to show self-recognition is also the age when they sometimes begin to act embarrassed (Lewis, Sullivan, Stanger, & Weiss, 1991). That is, they show a sense of self either in both situations or in neither.

The Preoperational Stage: Early Childhood

By about age 1½, most children are learning to speak; within a few years, they will have nearly mastered their language. Nevertheless, they do not understand everything in the same way that adults do. For example, they have difficulty

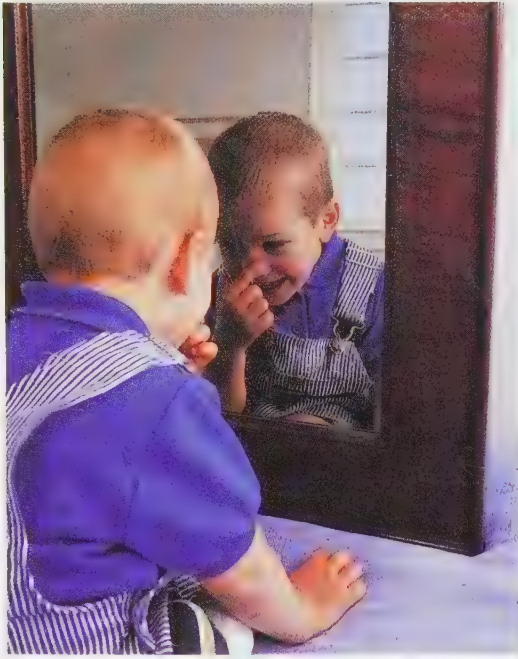


FIGURE 10.15 If someone places a bit of unscented rouge on a child's nose, a child about 2 years old or older shows self-recognition by touching his or her own nose. A younger child ignores the red spot or points at the mirror.

understanding that a mother can be someone else's daughter. A boy with one brother will assert that his brother has no brother. Piaget refers to this period as the **preoperational stage**, because the child lacks operations, which are reversible mental processes. For example, for a boy to understand that his brother has a brother, he must be able to reverse the concept of "having a brother."

Distinguishing Appearance from Reality in the Preoperational Stage Children in the early preoperational stage do not distinguish clearly between appearance and reality. A child who sees you put a white ball behind a blue filter will say that the ball is blue. When you ask, "Yes, I know the ball *looks* blue, but what color is it *really*?" the child grows confused. As far as the child is concerned, any ball that *looks* blue is blue (Flavell, 1986).

However, as we found in the case of object permanence, a child can appear to have an ability or to lack it, depending on how one tests that ability. Consider first the following experiment: A psychologist shows a child a playhouse room that is a scale model of a full-size room. Then, the psychologist hides a tiny toy in the small room (while the child watches) and explains that a bigger toy just like it is "in the same place" in the bigger room. (For example, if the little toy was behind the sofa in the little room, the big toy would be behind the sofa in the big room.) Then, the psychologist asks the child to go into the big room and find the big toy. Most 3-year-olds look in the correct place and find the toy immediately. Most 2½-year-olds, however, search haphazardly, instead of using the little room as a "map" for the big room. If the experimenter shows the child

the big toy in the big room and asks the child to find the little toy "in the same place" in the little room, the results are the same: 3-year-olds can find it but 2½-year-olds cannot. It appears that 3-year-olds have an ability that 2½-year-olds still lack, and this is in fact a striking example of a rapid intellectual change (DeLoache, 1989).

Before we speak too confidently about what a 2½-year-old cannot do, however, consider this clever follow-up study: The psychologist hides a toy in the small room while the child watches. Then, they step out of the room, and the psychologist shows the child a "machine that can expand things." The psychologist aims a beam from the machine at the room and takes the child out of the way. For the next couple of minutes they hear chunkata-chunkata-clunkata-clunkata sounds, and then the psychologist shows the child the full-sized room and asks the child to find the toy hidden in the blown-up room. Even 2½-year-olds have no trouble with this task; they go immediately to the correct location (DeLoache, Miller, & Rosengren, 1997). (See Figure 10.16.) Evidently, these children can use one room as a "map" of the other *if* they think of them both as "the same room," even though they don't seem to understand the idea that one room is a model of the other. (Incidentally, hardly any of the children showed any doubt that the machine had actually expanded the room, and many continued to believe it even after the psychologist tried to explain what had really happened!)

Egocentric Thinking in the Preoperational Period

Piaget concluded that children's thought processes are **egocentric**. In using this term, Piaget did *not* mean that children are selfish; instead, he meant that a child sees the world as centered around himself or herself and cannot take the perspective of another person. If you and a



Young children's thinking is egocentric: They have trouble understanding someone else's point of view. If asked to describe how a complicated pile of blocks appears to someone else, they instead describe how it appears from their own position.



a 2½-year-old shown in small room where stuffed animal is hidden



2½-year-old is unable to find the stuffed animal in the larger room



b 3-year-old shown in small room where stuffed animal is hidden



3-year-old is able to find the stuffed animal in the larger room

FIGURE 10.16 If an experimenter hides a small toy in a small room and asks a child to find the larger toy “in the same place” in the larger room, a 2½-year-old will search haphazardly. However, the same child knows exactly where to look, if the experimenter says this is the same room as before, except that a machine has expanded it.

preschool child sit on opposite sides of a complicated pile of blocks and you ask the child to draw what the blocks would look like from your side, the child will draw them as they look from his or her own side. When speaking, children often omit the necessary background information, as if assuming that the listener understands everything the speaker understands. (The same can be said for adults, unfortunately. Someone may start discussing the details of a topic before the listener has any idea what the speaker is talking about.)

CONCEPT CHECK

2. Which of the following is the clearest example of egocentric thinking?
 - a. A writer who uses someone else’s words without giving credit.
 - b. A politician who blames others for everything that goes wrong.
 - c. A professor who gives the same complicated lecture to a freshman class that she gives to a convention of professionals. (Check your answer on page 367.)

To say that a child is egocentric is to say that he or she has trouble understanding other people’s point of view—what they know and what they do not know. Some psycho-

logical researchers have explored the very difficult topic of what children understand about other people’s thoughts and knowledge.

WHAT IS THE EVIDENCE? *Children’s Understanding of Other People’s Cognitions*

How and when do children first understand that other people have minds and knowledge? Researchers have developed some very cleverly designed experiments to try to answer this difficult question.

Hypothesis A child who understands that other people have minds will distinguish between someone who is in a position to know certain information and someone who could not know it.

Method A 3- or 4-year-old child sat in front of four cups. (Figure 10.17.) The child watched as one adult hid a candy or toy under one of the cups, although a screen prevented the child from seeing which cup. Then, another adult entered the room. The “informed” adult pointed to the cup under which he or she had just hidden the surprise; the “uninformed” adult pointed to one of the other cups. The child was then given an opportunity to look under one cup to try to find the surprise.



FIGURE 10.17 A child sat in front of a screen covering four cups and watched as one adult hid a surprise under one of the cups. Then, that adult and another (who had not been present initially) each pointed to one of the cups to signal where the surprise was hidden. Many 4-year-olds consistently followed the advice of the informed adult; 3-year-olds did not.

This procedure was repeated 10 times for each child. The two adults alternated roles, but on each trial one or the other hid the surprise when the other was absent. That is, one was in a position to know where the surprise was hidden, and the other was not.

Results Of the 4-year-olds, 10 out of 20 chose the correct cup (the one indicated by the informed adult) at least 8 times out of 10 tries. That is, many of the 4-year-olds showed that they understood who had the relevant knowledge and who did not. However, none of 14 3-year-olds chose the correct cup as many as 8 times out of 10; they were just as likely to follow the lead of the uninformed adult as that of the informed adult (Povinelli & deBlois, 1992).

Interpretation Evidently, 4-year-olds have a greater understanding of other people's knowledge (or lack of it) than 3-year-olds have.

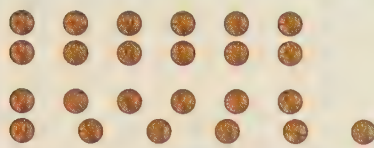
Other experiments, using a somewhat different procedure, have yielded similar results. For example, children in one study watched a dramatization where a girl who had a marble in her basket left the room temporarily. During her absence, a second girl moved the marble from the first girl's basket to her own basket. When the first girl returned to the room, the children were asked "Where is the marble?" and "Where will the girl look for it?" Most 4-year-olds answered that she would look in her own basket; younger children thought she would look in the other basket (Wimmer & Perner, 1983). As in the previous study, 4-year-olds are better able than younger children to make inferences about what various people might or might not know.

Although these are important results, we should avoid drawing too broad a conclusion. Using other methods, we can produce evidence that even younger children understand something about the experiences or knowledge of other people. For example, children less than 1 year old act sad and sometimes cry when they see another child get hurt (Hobson, 1993). That is, a child may show an understanding of other people's thoughts and feelings in some ways but not in others.

Lack of the Concept of Conservation in the Preoperational Period

According to Piaget, preoperational children lack the concept of conservation. Just as they fail to understand that something can still be white even though it looks blue, they fail to understand that objects conserve such properties as number, length, volume, area, and mass after the shape or arrangement of the objects has changed. They cannot perform the mental operations necessary to understand such transformations. (Table 10.1 shows some typical conservation tasks.)

For example, if we set up two glasses of the same size

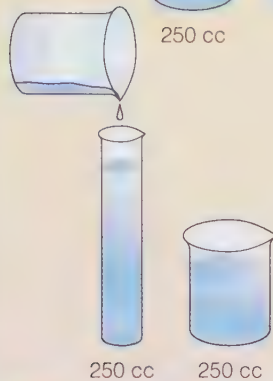
TABLE 10.1 Typical Tasks Used to Measure Conservation**Conservation of number**

Preoperational children say that these two rows contain the same number of pennies.

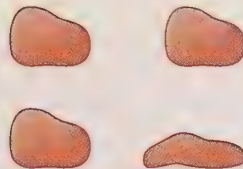
Preoperational children say that the second row has more pennies.

**Conservation of volume**

Preoperational children say that the two same-size containers have the same amount of water.



Preoperational children say that the taller, thinner container has more water.

**Conservation of mass**

Preoperational children say that the two same-size balls of clay have the same amount of clay.

Preoperational children say that a squashed ball of clay contains a different amount of clay than the same-size round ball of clay.

containing the same amount of water and then pour the contents of one glass into a taller, thinner glass, preoperational children will say that the second glass contains more water (Figure 10.18).

I once doubted whether children really believed what they were saying in such a situation. I thought perhaps

the phrasing of the questions somehow tricks them into saying something they do not believe. If you have these same doubts, borrow someone's 6-year-old child and try it yourself, with your own wording. Here's my own experience: One year when I was discussing Piaget in my introductory psychology

**TRY IT
YOUR-
SELF**

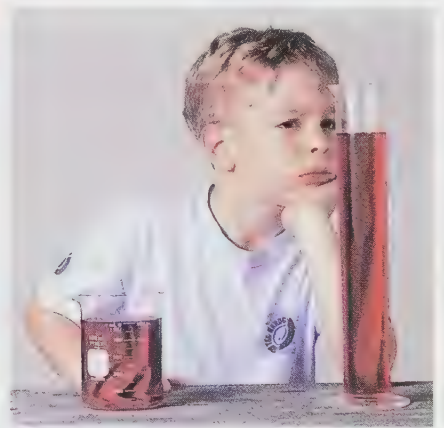
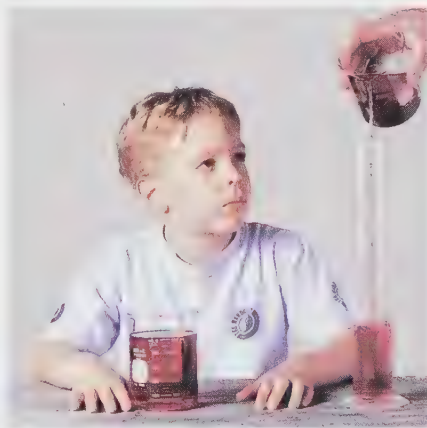


FIGURE 10.18 Looks can be deceptive: The conservation concept shows one way that children think less logically than adults do. Preoperational children, up to about the age of 7 years, don't understand that a property of a substance—such as the volume of water—remains constant despite changes in its appearance. During the transition from preoperational thinking to concrete operations at around age 7, conservation tasks seem difficult and confusing to the child.

class, I invited my son Sam, then 5½ years old, to take part in a class demonstration. I started with two glasses of water, which he agreed contained equal amounts of water. Then I poured the water from one glass into a wider glass, lowering the water level. When I asked Sam which glass contained more water, he confidently pointed to the tall, thin one.

After class he complained, “Daddy, why did you ask me such an easy question? Everyone could see that there was more water in that glass! You should have asked me something harder to show how smart I am!”

The following year, I brought Sam to class again for the same demonstration. He was now 6½ years old, about the age when children make the transition from preoperational thinking to the next stage. I again poured the water from one of the tall glasses into a wider one and asked him which glass contained more water. He looked and paused. His face got red. Finally, he whispered, “Daddy, I don’t know!” After class he complained, “Why did you ask me such a hard question? I’m never coming back to any of your classes again!” The question that was embarrassingly easy one year ago had, in the interim, become embarrassingly difficult.

The next year, when he was 7½, I tried again (at home). This time, he answered confidently, “Both glasses have the same amount of water, of course. Why? Is this some sort of trick question?”

The Concrete Operations Stage: Later Childhood

At about the age of 7, children enter the stage of concrete operations and begin to understand the conservation of physical properties. This transition is not sudden, however. The ability to understand the conservation of various properties emerges sequentially, at different ages. For instance, a 6-year-old child may understand that squashing a ball of clay will not change its weight but may not realize until years later that squashing the ball will not change the volume of water it displaces when it is dropped into a glass.

The **stage of concrete operations** is Piaget’s term for the stage when children can perform mental operations on concrete objects but still have trouble with abstract or hypothetical ideas. For example, ask this question: “How could you move a 4-mile-high mountain of whipped cream from one side of the city to the other?” Older children find the question amusing and try to think of an imaginative answer. But children in the concrete operations stage (or younger) are likely to complain that the question is silly.

Or ask, “If you could have a third eye anywhere on your body, where would you put it?” Children in this stage generally respond immediately that they would put it right between the other two, on their forehead. They seem to regard the question as uninteresting. Older children come up with more imaginative possibilities—on the back of their head or on the tip of a finger (so they could peek around corners).

The Formal Operations Stage: Adolescence and Adulthood

Formal operations is Piaget’s term for the mental processes used to deal with abstract, hypothetical situations. Those processes demand logical, deductive reasoning and systematic planning. According to Piaget, children reach the stage of formal operations at about age 11. He attributed some fairly sophisticated abilities to children in this stage, although later research indicates that many children take much longer to reach this stage, if they ever do.

Suppose we ask three children, ages 6, 10, and 14, to arrange a set of 12 sticks in order from the longest to the shortest. The 6-year-old (preoperational) child fails to order the sticks correctly. The 10-year-old (concrete operations) eventually gets them in the right order, but only after a great deal of trial and error. The 14-year-old (formal operations) holds the sticks upright with their bottom ends on the table and then removes the longest one, the second-longest one, and so on.

A second example: We set up five bottles of clear liquid and explain that it is possible, by mixing the liquids together in a certain combination, to produce a yellow liquid. The task is to find the right combination. Children in the concrete operations stage plunge right in with an unsystematic trial-and-error search. They try combining bottles A and B, then C and D, then perhaps A, C, and E, and so on. By the time they work through five or six combinations, they have forgotten which ones they’ve already tried. They may try one combination several times and others not at all; if and when they do stumble onto the correct combination, it is mostly by luck.

Children in the formal operations stage approach the problem more systematically. They may first try all the two-bottle combinations: AB, AC, AD, AE, BC, and so forth. If all those fail, they then try three-bottle combinations: ABC, ABD, ABE, ACD, and so on. By adopting a strategy for trying every possible combination only once, they are bound to succeed.

Children do not reach the stage of formal operations any more suddenly than they reach the concrete operations stage. Before they can reason logically about a particular problem, they must first have a fair amount of experience in dealing with that problem. A 9-year-old who has spent a great deal of time playing chess reasons logically about chess problems and plans several moves ahead. The same child reverts to concrete reasoning, however, when faced with an unfamiliar problem.

Table 10.2 summarizes Piaget’s four stages.

CONCEPT CHECK

3. You are given the following information about four children. Assign each of them to one of Piaget’s stages of intellectual development. (Check your answers on page 367.)
 - a. Child has mastered the concept of conservation; still has trouble with abstract and hypothetical questions.